Cambridge (CIE) IGCSE Additional Maths

Your notes

Calculus for Kinematics

Contents

Kinematics Toolkit

Your notes

Displacement, Velocity & Acceleration

What is kinematics?

- **Kinematics** is the branch of mathematics that models and analyses the **motion** of objects
- Common words such as **distance, speed** and **acceleration** are used in kinematics but are used according to their technical definition

What definitions do I need to be aware of?

- Firstly, only motion of an object in a **straight line** is considered
  - this could be a **horizontal** straight line
    - the **positive** direction would be to the **right**
  - or this could be a **vertical** straight line
    - the **positive** direction would be **upwards**
- **Particle**
  - A **particle** is the general term for an **object**
  - some questions may use a **specific** object such as a **car** or a **ball**
- **Time** t seconds
  - **Displacement**, **velocity** and **acceleration** are all **functions** of **time** t
  - **Initially** time is zero $t = 0$
- **Displacement** s m
  - The **displacement** of a particle is its **distance relative** to a **fixed point**
  - the fixed point is often (but not always) the particle's **initial position**
  - **Displacement** will be **zero** $s = 0$ if the object is at or has returned to its initial position
  - **Displacement** will be negative if its **position relative** to the **fixed point** is in the **negative direction** (left or down)
- **Distance** d m

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- Use of the word **distance** needs to be considered carefully and could refer to
  - the distance **travelled** by a particle
  - the **(straight line)** distance the particle is from a **particular point**
- Be careful not to confuse **displacement** with **distance**
- if a bus route starts and ends at a bus depot, when the bus has returned to the depot, its **displacement** will be **zero** but the distance the bus has travelled will be the length of the route
- **Distance** is always **positive**

- **Velocity** v m s^{-1}
  - The **velocity** of a particle is the **rate of change** of its **displacement** at time t
  - **Velocity** will be **negative** if the **particle** is moving in the **negative direction**
  - A **velocity** of **zero** means the particle is **stationary** $v = 0$
- **Speed** $|v|$ m s^{-1}
  - **Speed** is the **magnitude** (a.k.a. absolute value or modulus) of **velocity**
  - as the particle is **moving** in a **straight line, speed** is the **velocity ignoring** the **direction**
    - if $v = 4,\ |v| = 4$
    - if $v = -6,\ |v| = 6$
- **Acceleration** a m s^{-2}
  - The **acceleration** of a particle is the **rate of change** of its **velocity** at time t
  - Acceleration can be **negative** but this alone cannot fully describe the particle's motion
    - if **velocity** and **acceleration** have the **same** sign the particle is **accelerating** (speeding up)
    - if **velocity** and **acceleration** have **different** signs then the particle is **decelerating** (slowing down)
    - if **acceleration** is **zero** $a = 0$ the particle is moving with **constant** velocity
    - in all cases the **direction** of **motion** is determined by the **sign** of **velocity**

Are there any other words or phrases in kinematics I should know?

- Certain words and phrases can imply values or directions in kinematics

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- a particle described as "at **rest**" means that its velocity is zero, $v = 0$
- a particle described as moving "**due east**" or "**right**" or would be moving in the **positive horizontal** direction
  - this also means that $v > 0$
- a particle "**dropped from the top of a cliff**" or "**down**" would be moving in the **negative vertical** direction
  - this also means that $v < 0$

What are the key features of a velocity-time graph?

- The **gradient** of the graph equals the **acceleration** of an object
- A **straight line** shows that the object is **accelerating** at a **constant rate**
- A **horizontal** line shows that the object is moving at a **constant velocity**
- Graph **above** the x-axis means the object is moving **forwards**
  - Graph **below** the x-axis means the object is moving **backwards**The **area** between graph and the x-axis tells us the **change in displacement** of the object
- The **total displacement** of the object from its starting point is the sum of the **areas above** the x-axis **minus** the sum of the **areas below** the x-axis
- The **total distance travelled** by the object is the sum of **all** the **areas**
- If the graph **touches** the **x-axis** then the object is **stationary** at that time
- If the graph is **above** the **x-axis** then the object has positive velocity and is **travelling forwards**
- If the graph is **below** the **x-axis** then the object has negative velocity and is **travelling backwards**

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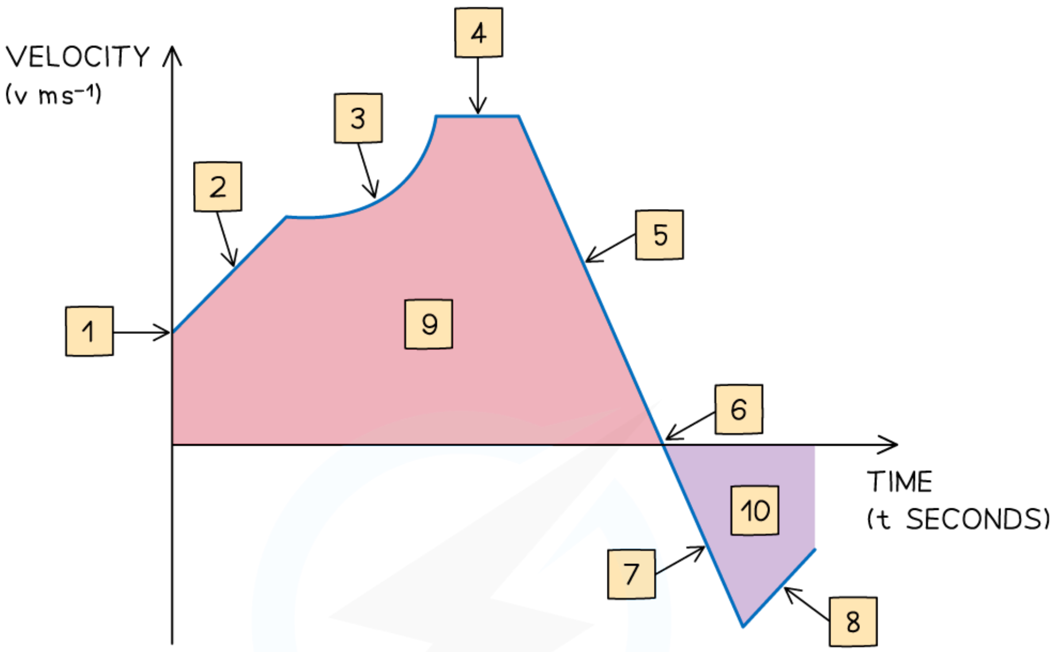

1. INITIAL VELOCITY
2. CONSTANT ACCELERATION
3. VARIABLE ACCELERATION
4. CONSTANT VELOCITY
5. DECELERATING (SLOWING DOWN BUT STILL MOVING FORWARDS)
6. INSTANTANEOUSLY AT REST (STATIONARY FOR AN INSTANT)
7. SPEEDING UP BUT MOVING BACKWARDS
8. SLOWING DOWN BUT STILL MOVING BACKWARDS
9. DISTANCE TRAVELLED FORWARDS
10. DISTANCE TRAVELLED BACKWARDS

Examiner Tips and Tricks

- In an exam if you are given an expression for the velocity then sketching a velocity-time graph can help visualise the problem

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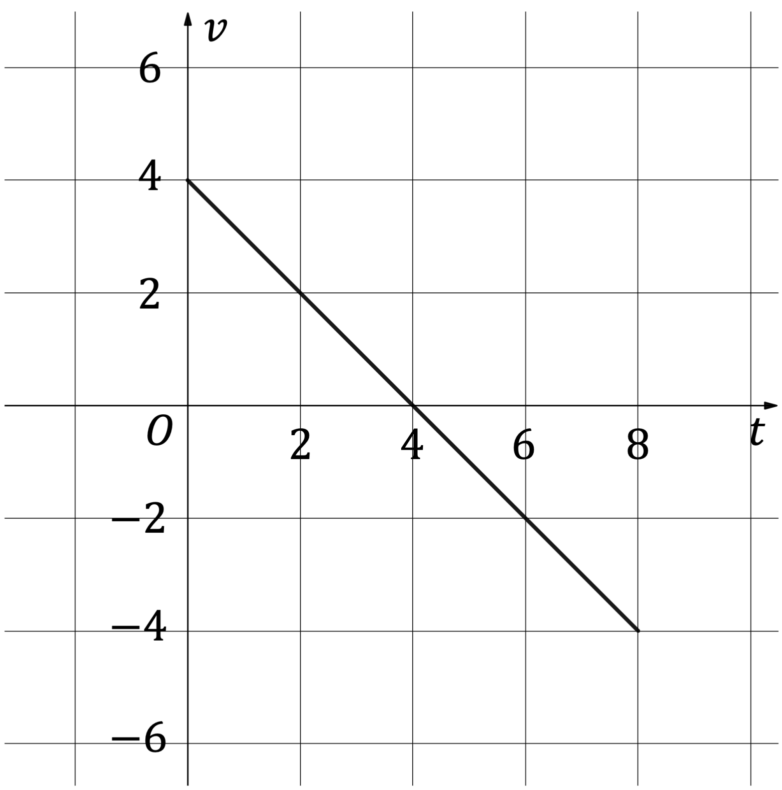
Worked Example

A particle is projected vertically upwards from ground level, taking 8 seconds to return to the ground.

The velocity-time graph below illustrates the motion of the particle for these 8 seconds.



i) How many seconds does the particle take to reach its maximum height? Give a reason for your answer.

ii) State, with a reason, whether the particle is accelerating or decelerating at time $t = 3$.

i) At maximum height, velocity is zero.

$v = 0$ at $t = 4$

The particle takes **4 seconds** to reach its maximum height. This is because its **velocity is 0 ms^{-1}** at 4 seconds.

ii)

At $t = 3$, velocity is positive

Acceleration is the gradient of velocity.

At $t = 3$, acceleration is negative

At 3 seconds the particle is **decelerating** as its **velocity and acceleration have different signs**.

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Calculus for Kinematics

Differentiation for Kinematics

How is differentiation used in kinematics?

- **Displacement**, **velocity** and **acceleration** are related by calculus
- In terms of differentiation and derivatives
  - **velocity** is the **rate** of **change** of **displacement**
    - $v = \frac{\mathrm{d}s}{\mathrm{d}t}$ or $v(t) = s'(t)$
  - **acceleration** is the **rate** of **change** of **velocity**
    - $a = \frac{\mathrm{d}v}{\mathrm{d}t}$ or $a(t) = v'(t)$
  - so **acceleration** is also the **second derivative** of **displacement**
    - $a = \frac{\mathrm{d}^2 s}{\mathrm{d}t^2}$ or $a(t) = s''(t)$

Worked Example

The displacement, x m, of a particle at t seconds, is modelled by the function $x(t) = 2t^3 - 27t^2 + 84t$.

Find expressions for $x'(t)$ and $x''(t)$, and state what these expressions represent.

$$x = 2t^3 - 27t^2 + 84t$$

$x'(t)$ means the same as $\frac{\mathrm{d}x}{\mathrm{d}t}$

$$\begin{aligned} x'(t) &= 6t^2 - 54t + 84 \\ x'(t) &= 6(t^2 - 9t + 14) \end{aligned}$$

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$$x'(t) = 6(t-2)(t-7)$$

This expression represents the velocity of the particle.

Answers may or may not need to be factorised, depending on the question

$x''(t)$ means the same as $\frac{\mathrm{d}^2x}{\mathrm{d}t^2}$

$$x''(t) = 12t - 54$$

$$x''(t) = 6(2t-9)$$

This expression represents the acceleration of the particle.

Integration for Kinematics

How is integration used in kinematics?

- Since **velocity** is the **derivative** of **displacement** ($v = \frac{\mathrm{d}s}{\mathrm{d}t}$) it follows that

$$s = \int v \, \mathrm{d}t$$

- Similarly, **velocity** will be an **antiderivative** of **acceleration**

$$v = \int a \, \mathrm{d}t$$

How would I find the constant of integration in kinematics problems?

- A **boundary** or **initial** condition would need to be known
  - phrases involving the word **"initial"**, or **"initially"** are referring to **time** being **zero**, i.e. $t = 0$
  - you might also be given information about the object at some other time (this is called a **boundary condition**)
  - **substituting** the values in from the **initial or boundary condition** would allow the **constant** of **integration** to be found

How are definite integrals used in kinematics?

- Definite integrals can be used to find the displacement of a particle between two points in time

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- $\int_{t_1}^{t_2} v(t)\, \mathrm{d}t$ would give the **displacement** of the particle **between** the times $t = t_1$ and $t = t_2$
  - This can be found using a velocity-time graph by **subtracting** the **total area below** the horizontal axis from the **total area above**
  - You could think of this as the "net area" e.g. 4 m - 1 m = 3 m
- $\int_{t_1}^{t_2} |v(t)|\, \mathrm{d}t$ gives the **distance** a particle has **travelled** between the times $t = t_1$ and $t = t_2$
  - This can be found using a velocity velocity-time graph by **adding** the **total area below** the horizontal axis to the **total area above**
  - You could think of this as the "gross area" e.g. 4 m + 1 m = 5 m

Examiner Tips and Tricks

- Sketching the velocity-time graph can help you visualise the distances travelled using areas between the graph and the horizontal axis

Worked Example

A particle moving in a straight horizontal line has velocity v ms^{-1} at time t seconds modelled by $v(t) = 8t^3 - 12t^2 - 2t$.

a) Given that the initial position of the particle is at the origin, find an expression for its displacement from the origin at time t seconds.

The integral of the velocity gives the displacement

$$s(t) = \int v(t)\ \mathrm{d}t = \int (8t^3 - 12t^2 - 2t)\ \mathrm{d}t$$

$$s(t) = 2t^4 - 4t^3 - t^2 + c$$

"Initial" means when $t = 0$ and "at the origin" means $s = 0$

Substitute these values in to find c

$$0 = 2(0)^4 - 4(0)^3 - (0)^2 + c$$
$$c = 0$$

$$\boldsymbol{s(\mathrm{t}) = 2t^4 - 4t^3 - t^2}$$

b) Find the displacement of the particle from the origin after the first five seconds of its motion.

To find the displacement after 5 seconds, integrate the velocity between 0 and 5

$$\int_0^5 8t^3 - 12t^2 - 2t \, \mathrm{d}t$$
$$[2t^4 - 4t^3 - t^2]_0^5$$
$$= [2(5)^4 - 4(5)^3 - (5)^2] - [2(0)^4 - 4(0)^3 - (0)^2]$$
$$= 725 - 0$$

725 m

c) Explain why this value is not the same as the distance travelled during the first 5 seconds.

When finding the displacement, any negative displacements are taken into account. e.g. 3m - 1m = 2m. When finding the distance, we are looking for the total of all the areas, ignoring their direction/sign. e.g. 3m + 1m = 4m. This function for velocity has some regions underneath the x-axis, and some regions above the x-axis between 0 and 5 seconds, so the displacement and distance covered in this time will be different.

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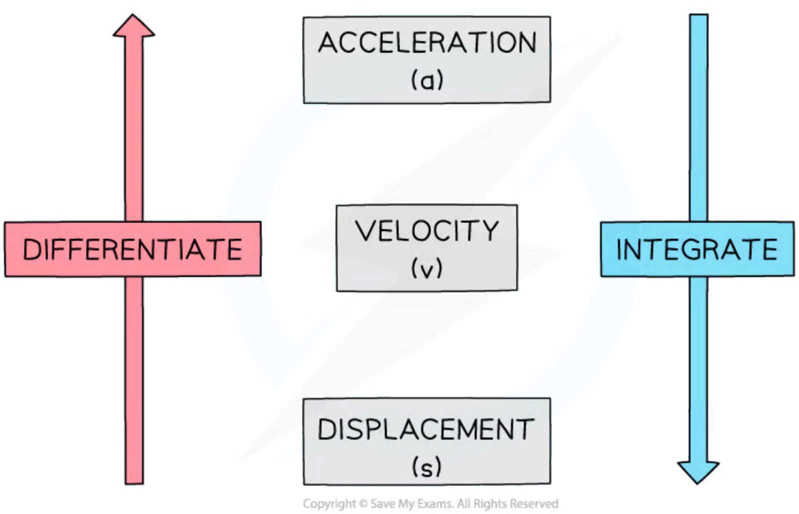
Sketching Travel Graphs

Your notes

Sketching Travel Graphs

How are s-t, v-t, and a-t graphs related?

- Recall that:
  - Velocity, **v**, is the rate of change of displacement, **s**, with respect to time
  - Acceleration, **a**, is the rate of change of velocity, **v**, with respect to time
  - **Differentiate** to go from **s** to **v** and from **v** to **a**
  - **Integrate** to go from **a** to **v** and from **v** to **s**
    - There will be a **constant of integration**, c, **each time** you integrate



- On a **velocity-time graph:**
  - **Acceleration** is the **gradient** which is found using **differentiation**
  - **Displacement** is the **area** under the graph which is found using **integration**
- This can also be seen from the units:
  - $\text{Gradient} = \dfrac{\text{ms}^{-1}}{\text{s}} = \text{ms}^{-2} = \text{Acceleration}$

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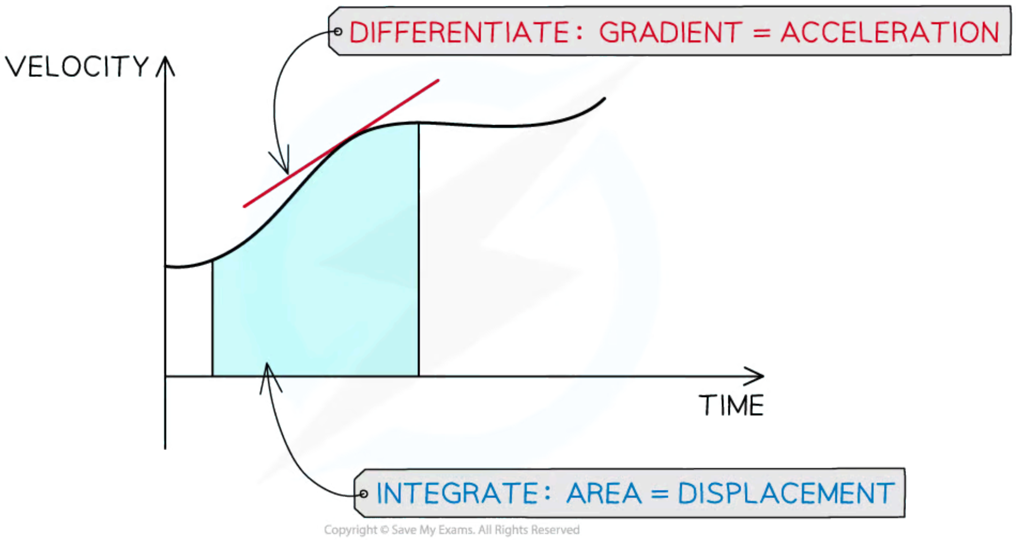
- $\text{Area } = \text{ms}^{-1} \times \text{s } = \text{m} = \text{Displacement}$



- On a **displacement-time graph:**
  - **Velocity** is the **gradient** which is found using **differentiation**
  - The area has no significant meaning
- This can also be seen from the units:
  - $\text{Gradient } = \frac{\text{m}}{\text{s}} = \text{ms}^{-1} = \text{Velocity}$
- On an **acceleration-time graph:**
  - **Velocity** is the **area** under the graph which is found using **integration**
  - The gradient is generally not used
    - It is a measure called 'jolt' but this is beyond the scope of this course
- This can also be seen from the units:
  - $\text{Area } = \text{ms}^{-2} \times \text{s } = \text{ms}^{-1} = \text{Velocity}$

How can I use one travel graph to draw another?

- Using the relations stated above, we can inspect either the graph or the equation of the graph, in order to sketch a related travel graph

Your notes

- For example, if a **velocity-time** graph is a **series of sections**, with mostly straight lines:
  - Find the **gradient** of each section to plot the **acceleration-time** graph
    - Remember if the gradient is negative, the acceleration-time graph will be below the x-axis
  - Find the **area** underneath each section to help plot the **displacement-time** graph
    - Remember that if the velocity is a positive constant (a horizontal line above the x-axis), the displacement will be increasing (a line with positive gradient)
    - If the velocity is a negative constant (a horizontal line below the x-axis), the displacement will be decreasing (a line with negative gradient)
- If a graph is a **curve with a known equation**, we can use **calculus** to find the equations of the other related functions
  - Remember you may need extra information about the velocity or displacement at a point in time when integrating
  - Once the equations of the other functions are found, they can be **sketched**
  - For example if the graph of the **displacement**-time graph is a **cubic**
    - the **velocity**-time graph will be a **quadratic** graph
    - and the **acceleration**-time graph will be a **linear** graph

Corresponding s-t, v-t, and a-t graphs for the same journey

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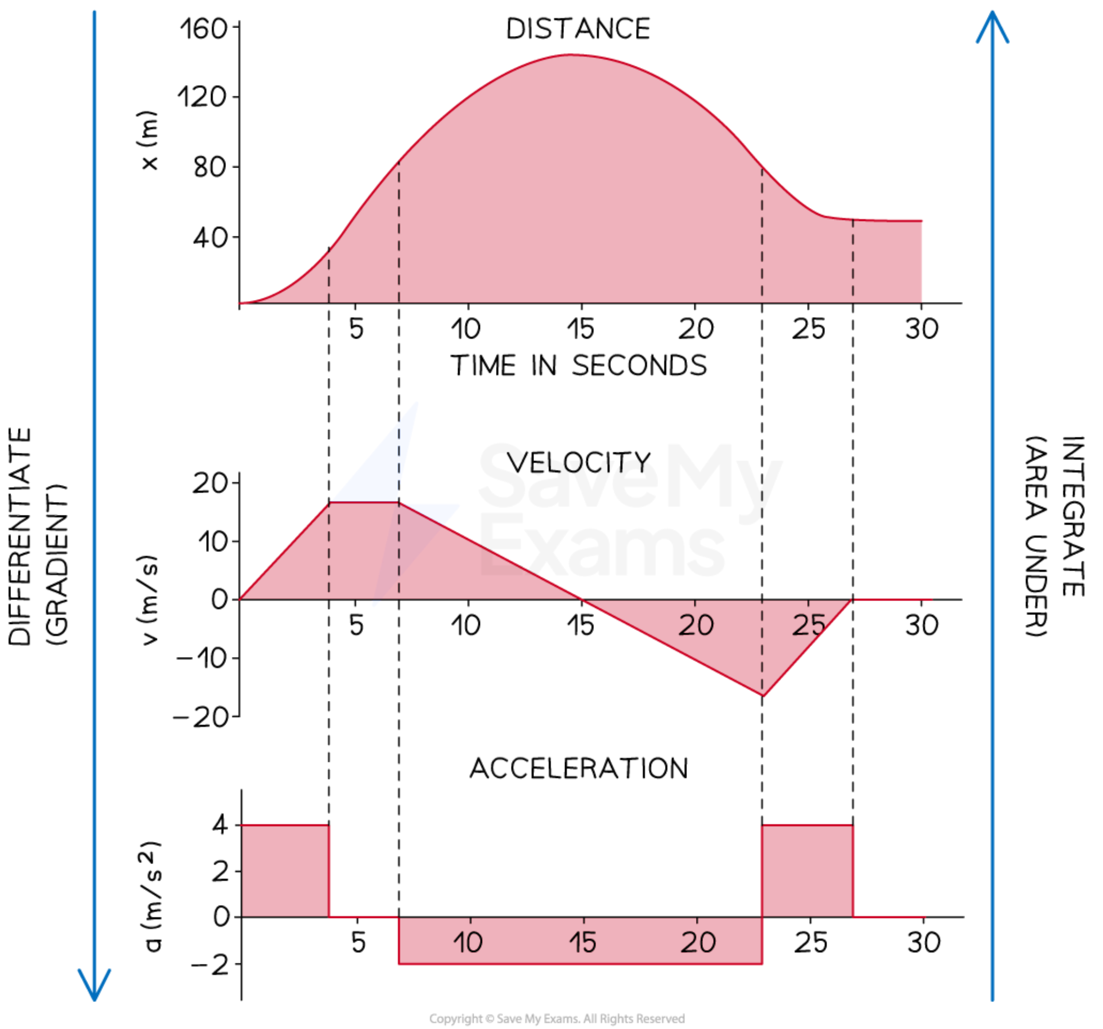

Examiner Tips and Tricks

- Questions may involve both differentiation and integration (or finding gradients and areas)
  - take a moment to double check you have selected the correct method!

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Worked Example

A particle moves in a straight line. Its displacement, s metres, from a fixed point at time, t seconds, is given by $s = -2t^3 + 12t^2$ for $0 \leq t \leq 7$.

Sketch its displacement-time, velocity-time, and acceleration-time graphs.

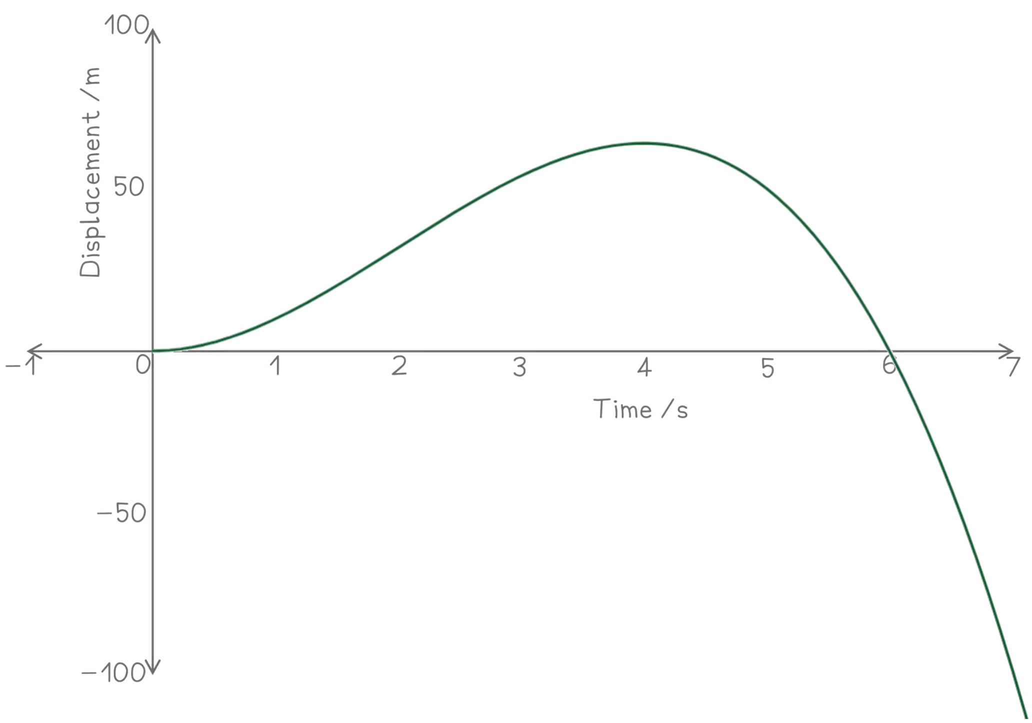
To sketch the displacement; $s = -2t^3 + 12t^2$ it can be factorised

$$s = -2t^3 + 12t^2 = -2t^2(t-6)$$

The roots can then be found

$s = 0$ when:
$t = 0$ (repeated root)
and $t = 6$

The graph can then be sketched, noting that it is a negative cubic, and remembering the restriction on the domain; $0 \leq t \leq 7$



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To find the velocity, differentiate the displacement with respect to t (time)

$$v = \frac{\mathrm{d}s}{\mathrm{d}t} = -6t^2 + 24t$$

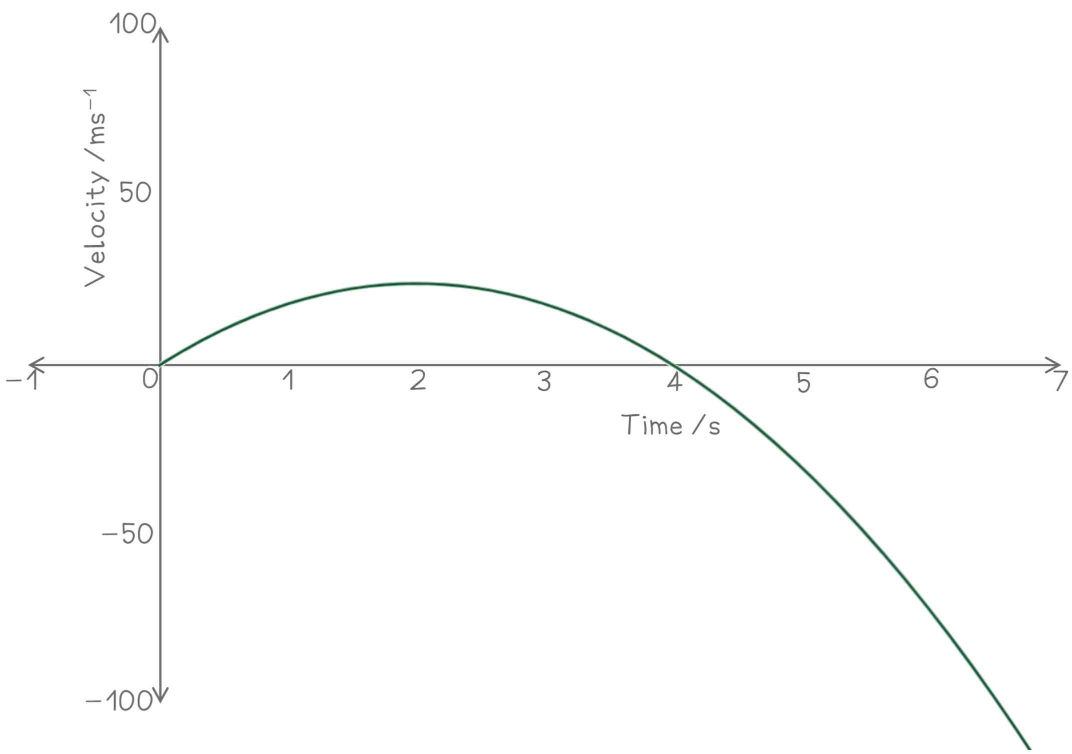
This can be factorised to

$$v = -6t(t-4)$$

The roots can then be found

$$v = 0 \text{ when:}$$
$$t = 0 \text{ and } t = 4$$

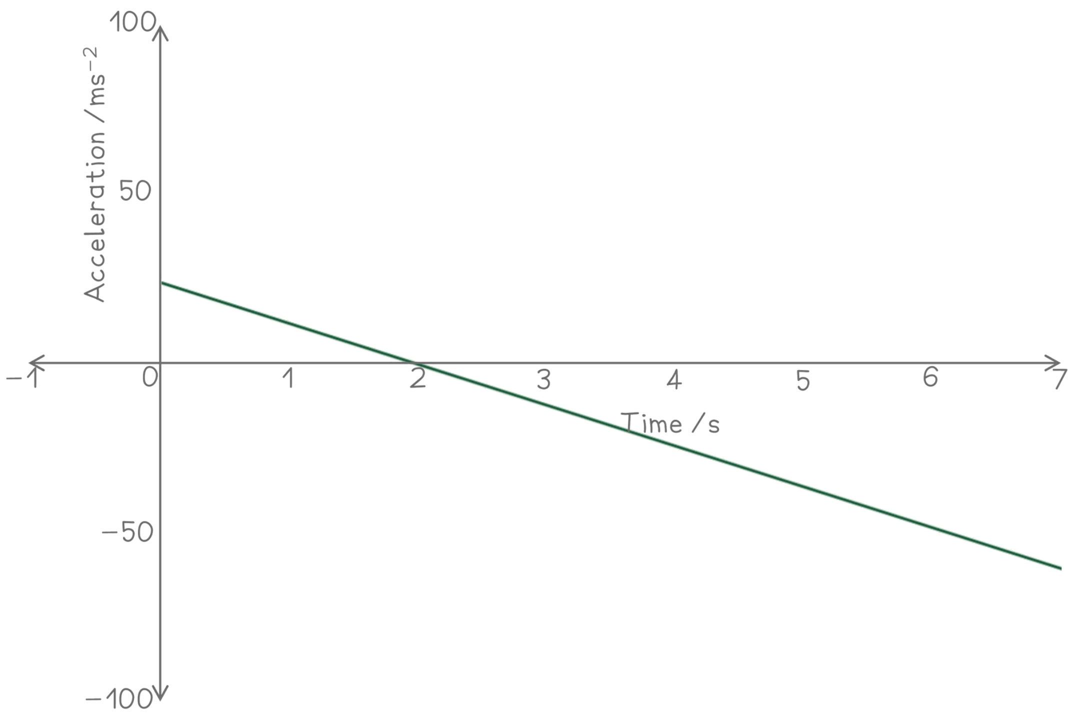
The graph can then be sketched, noting that it is a negative quadratic, and remembering the restriction on the domain; $0 \le t \le 7$



To find the acceleration, differentiate the velocity with respect to t This is also the second derivative of the displacement

$$a = \frac{d^2 s}{dt^2} = \frac{dv}{dt} = -12t + 24$$

The graph can then be sketched This is a straight line with y-intercept 24, and gradient -12
Remember the restriction on the domain; $0 \le t \le 7$



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